

Symbiotic Relay Audit Summary

Prepared by Cyfrin

Version 2.1

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September 15, 2025

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Symbiotic on Symbiotic Relay. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.1 dated September 15, 2025, was delivered to Symbiotic, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 15 days, the Cyfrin team conducted an audit on the Symbiotic Relay code provided by Symbiotic. The findings consist of 3 Medium & 2 Low severity issues with the remainder being informational and gas optimizations.

Summary

Project Name	Symbiotic Relay
Repository	relay-contracts
Commit	1a1f94fa1798...
Fix Commit	9759b26a8364...
Audit Timeline	July 15th - August 4th, 2025
Methods	Manual Review

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	0	0	0
Medium Risk	3	1	2
Low Risk	2	2	0
Informational	5	1	4
Gas Optimizations	3	1	2
Total	13	5	8

6 Audit Scope

The Symbiotic Middleware SDK is a comprehensive framework that enables the development of custom middleware solutions for restaking protocols. The scope excludes Symbiotic Core contracts and focuses exclusively on the middleware SDK components that enable custom middleware development and deployment within the Symbiotic ecosystem.

All files in the `src/**` folders were part of the audit scope except for the `BN254.sol`.

The scope of this audit was limited to:

```
relay-contracts/src/contracts/libraries/keys/KeyBlsBn254.sol
relay-contracts/src/contracts/libraries/keys/KeyEcdsaSecp256k1.sol
relay-contracts/src/contracts/libraries/sigs/SigBlsBn254.sol
relay-contracts/src/contracts/libraries/sigs/SigEcdsaSecp256k1.sol
relay-contracts/src/contracts/libraries/structs/Checkpoints.sol
relay-contracts/src/contracts/libraries/structs/PersistentSet.sol
relay-contracts/src/contracts/libraries/utils/InputNormalizer.sol
relay-contracts/src/contracts/libraries/utils/KeyTags.sol
relay-contracts/src/contracts/libraries/utils/Scaler.sol
relay-contracts/src/contracts/libraries/utils/ValSetVerifier.sol
relay-contracts/src/contracts/modules/base/NetworkManager.sol
relay-contracts/src/contracts/modules/base/OzEIP712.sol
```

```
relay-contracts/src/contracts/modules/base/PermissionManager.sol
relay-contracts/src/contracts/modules/common/permissions/OzAccessControl.sol
relay-contracts/src/contracts/modules/common/permissions/OzAccessManaged.sol
relay-contracts/src/contracts/modules/common/permissions/OzOwnable.sol
relay-contracts/src/contracts/modules/key-registry/KeyRegistry.sol
relay-contracts/src/contracts/modules/network/Network.sol
relay-contracts/src/contracts/modules/settlement/sig-verifiers/libraries/ExtraDataStorageHelper.sol
relay-contracts/src/contracts/modules/settlement/sig-verifiers/SigVerifierBlsBn254Simple.sol
relay-contracts/src/contracts/modules/settlement/sig-verifiers/SigVerifierBlsBn254ZK.sol
relay-contracts/src/contracts/modules/settlement/Settlement.sol
relay-contracts/src/contracts/modules/valset-driver/EpochManager.sol
relay-contracts/src/contracts/modules/valset-driver/ValSetDriver.sol
relay-contracts/src/contracts/modules/voting-power/base/VotingPowerCalcManager.sol
relay-contracts/src/contracts/modules/voting-power/common/voting-power-calc/libraries/ChainlinkPriceFee
↪ d.sol
relay-contracts/src/contracts/modules/voting-power/common/voting-power-calc/EqualStakeVPCalc.sol
relay-contracts/src/contracts/modules/voting-power/common/voting-power-calc/NormalizedTokenDecimalsVPCa
↪ lc.sol
relay-contracts/src/contracts/modules/voting-power/common/voting-power-calc/PricedTokensChainlinkVPCalc
↪ .sol
relay-contracts/src/contracts/modules/voting-power/common/voting-power-calc/WeightedTokensVPCalc.sol
relay-contracts/src/contracts/modules/voting-power/common/voting-power-calc/WeightedVaultsVPCalc.sol
relay-contracts/src/contracts/modules/voting-power/extensions/logic/BaseRewardsLogic.sol
relay-contracts/src/contracts/modules/voting-power/extensions/logic/BaseSlashingLogic.sol
relay-contracts/src/contracts/modules/voting-power/extensions/logic/OpNetVaultAutoDeployLogic.sol
relay-contracts/src/contracts/modules/voting-power/extensions/BaseRewards.sol
relay-contracts/src/contracts/modules/voting-power/extensions/BaseSlashing.sol
relay-contracts/src/contracts/modules/voting-power/extensions/MultiToken.sol
relay-contracts/src/contracts/modules/voting-power/extensions/OperatorsBlacklist.sol
relay-contracts/src/contracts/modules/voting-power/extensions/OperatorsJail.sol
relay-contracts/src/contracts/modules/voting-power/extensions/OperatorsWhitelist.sol
relay-contracts/src/contracts/modules/voting-power/extensions/OperatorVaults.sol
relay-contracts/src/contracts/modules/voting-power/extensions/OpNetVaultAutoDeploy.sol
relay-contracts/src/contracts/modules/voting-power/extensions/SharedVaults.sol
relay-contracts/src/contracts/modules/voting-power/logic/VotingPowerProviderLogic.sol
relay-contracts/src/contracts/modules/voting-power/VotingPowerProvider.sol
```

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

Symbiotic SDK is a comprehensive middleware infrastructure that enables permissionless validator management and signature aggregation for decentralized networks. The system is designed to collect and aggregate signatures from validators, maintain validator sets, and facilitate settlement of network state.

9 Attestation

Cyfrin attests that it performed a security review of the Symbiotic Relay code provided by Symbiotic.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2025-09-04-cyfrin-symbiotic-v2.1.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Symbiotic has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=a89b508b9780812cd194ea3cdbb1977485636aa7a850be90420d59d7b9a912b5
F=2025-09-04-cyfrin-symbiotic-v2.1-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.